Problem Summary

Given an array of heights, where each value represents the height of a vertical line, find 2 lines that together form a container holding the most water.

The amount of water is: width x height, where

→ width = distance between the two lines

→ height = Shorter of the two lines

Key Idea

Use 2 pointers:

→ left starts at beginning

→ right starts at the end

At each step:

→ calculate the area

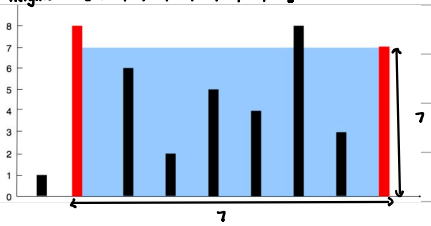
→ update the maximum area

→ move the pointer with a smaller height inward

* Shorter line limits the height

Diagram Examples

heights = [1, 8, 6, 2, 5, 4, 8, 3, 7]



Area = 7 x 7 = 49

Approach

1) Set left pointer at start

2) Set right pointer at end

3) While left < right:

→ Compute current area

→ update max_area

→ move the pointer with smaller height

4) Return max_area

Time Complexity: O(n) → each pointer moves at most once across the array

Space Complexity: O(1) → only pointers are used